Anuj Pratap Singh Yadav

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SUMMARY

Staff engineer, 11 years of C++, the last three-plus in formal verification at Synopsys. I own coverage for a C++ vs RTL equivalence app, lead cross-team architecture changes, and build LLVM analysis of C++ designs and AI agents.

SKILLS

Formal verification: VC Formal (DPV: C++ vs RTL equivalence; FPV), formal and code coverage, property density, cone of influence, sign-off

Languages: C++, Tcl, Python, C#, Java, TypeScript

Tools: Verdi, LLVM IR passes (dg), LSF, FlexNet, GDB, Linux, Perforce, Git, Win32

C++ architecture: dependency inversion, IWYU, serialization (XML, pugixml)

AI tooling: Cursor agents, skills, rules and subagents

EXPERIENCE

R&D Staff Engineer | Synopsys *Noida, India | May 2023 - Present*

VC Formal (formal verification), DPV app: C++ vs RTL equivalence

- **C++ code coverage for C++ vs RTL sign-off**: own it end to end. It shows which parts of the C++ reference model the formal proofs actually exercised, so the C++ side can be signed off like RTL. Added merging across proofs (covered if any proof covers a block, unreachable only if all prove it) and export to the standard coverage database, so C++ and RTL coverage are reviewed together in Verdi; handle customer issues and guided its integration into a new C++ + RTL sign-off app.
- **Formal coverage (property density, RTL coverage)**: own both. They report which design signals, down to the bit, and which RTL lines the properties actually check. Rebuilt property density so DPV and FPV give the same answer, adding bit-level results, RTL-to-RTL designs and a PASS/FAIL output-coverage check (beta); aligned RTL coverage semantics across the two apps. In production.
- **C++ cone of influence (R&D)**: building source-level cone of influence for C++ designs, which reports the exact C++ lines a property depends on. Wrote LLVM IR passes on the open-source dg slicer for C++ constructs it missed, and a signal-based slicing criterion that cut whole-file slices down to the lines feeding each signal. Design reviewed weekly with a Distinguished Engineer.
- **Build-dependency decoupling (multi-org initiative)**: led VC Formal's part. The shared C++ layer under ~15 apps and tools included engine headers, so one engine edit rebuilt every app. Cut wrong-way dependencies ~73% with include-what-you-use, forward declarations, dependency inversion through startup-injected callbacks, and code relocation, in small changes reviewed by the owning teams; in production. Org-wide tech talk on the design.
- **Licensing for distributed proofs**: built token licensing in the C++ proof engine and fixed production license-checkout failures in the master/worker distributed solver.
- **Customer escalations**: own them end to end in my areas: root cause, regression test, fix shipped in service packs.
- **AI agents for the team's workflow**: built Cursor agents teammates use: a daily Jira check that flags tickets with wrong validation fields and notifies owners, and an agent that reproduces a Jira issue, root-causes it and drafts a fix as a code review for an engineer to check.
- **Session save/restore**: closed silent state-loss gaps (user Tcl procedures, engine state as XML), with round-trip tests.

Software Engineer II | Microsoft *Noida, India | May 2021 - May 2023*

- Owned Recording Studio features in PowerPoint Win32 (C++): camera modes, live background blur, video export and Cameo, a camera object placed on the slide. Led v1 test automation and wired Recording and Export telemetry.
- Designed catalog filtering plus an allowlist (C#) so new commerce SKUs no longer broke OneDrive's upgrade path.
- Built the Stream 2.0 video playback component in Excel Win32 (C++, TypeScript) for SharePoint and OneDrive video.

Software Development Engineer 2 | Adobe *Noida, India | Aug 2015 - Apr 2021*

- Moved Photoshop and Premiere Elements from RIBS to Hyperdrive builds with parallel bundles, cutting install time ~50%.
- Designed lazy loading for large files in the AEM web editor, cutting load time ~60% on files customers reported as freezing.

EDUCATION AND AWARDS

Spot Awards, Synopsys (latest Sep 2026, org all-hands) | Spot Award, Adobe (RIBS to Hyperdrive migration)

B.Tech, Computer Science and Engineering - NIT Bhopal (MANIT) | 2011 - 2015 | GPA 8.27/10